

Implement Client Script & UI Policy (Incident)

Phase 2: Requirement Analysis – Solution Requirements

Item	Details
Project Title	Implement Client Script & UI Policy (Incident)
Platform	ServiceNow
Target Table	Incident [incident]
Sub-topic	Solution Requirements (Functional and Non-Functional)
Team ID	SWTID-2026-6974

1. Introduction

This section translates the problem statement from the Ideation Phase into clear requirements. It states what the solution must do (functional requirements), how well it must do it (non-functional requirements), and the business rules, data elements, assumptions and constraints that shape the implementation.

2. Requirement Overview

Incident records must be complete and consistent before they are submitted. The solution enforces this at the user interface level using one UI Policy, one UI Policy Action and three Client Scripts on the Incident table.

2.1 Business Rules

ID	Business Rule
BR1	When Impact is 1 - High, an assignment group must be provided.
BR2	When Impact is 1 - High, Urgency must be High and cannot be edited by the user.
BR3	When Impact is 1 - High, an incident cannot be saved unless Assigned to is filled in.
BR4	When Impact is no longer High, the field restrictions applied by the UI Policy are reversed.
BR5	State cannot be changed by editing directly in the list view; it must be changed on the Incident form.

3. Functional Requirements

ID	Requirement	Description	Priority	Rule
FR1	High Impact Control policy	Create an active UI Policy on the Incident table with the condition Impact is 1 - High.	High	BR1
FR2	Mandatory Assignment group	A UI Policy Action on Assignment group sets Mandatory to true when the condition is met.	High	BR1
FR3	Reverse if false	The UI Policy reverses its actions when Impact is not High, so fields return to normal.	High	BR4
FR4	Read-only Urgency	A UI Policy Action on Urgency sets Read-only to true and leaves Visible unchanged.	High	BR2

ID	Requirement	Description	Priority	Rule
FR5	Auto-set Urgency	An onChange Client Script on Impact sets Urgency to 1 when Impact changes to High and shows an information message.	High	BR2
FR6	Ignore form load	The onChange script must not run while the form is loading or when the new value is empty.	Medium	BR2
FR7	Save validation	An onSubmit Client Script blocks the save when Impact is High and Assigned to is empty.	High	BR3
FR8	Error message	The onSubmit script shows an error on Assigned to: "Assigned To is mandatory for High impact incidents."	High	BR3
FR9	List edit blocking	An onCellEdit Client Script on State shows an alert and cancels the edit.	Medium	BR5
FR10	Form-based State change	Users must still be able to change State on the Incident form and save successfully.	Medium	BR5
FR11	Successful save	An incident with valid data, including Assigned to for High impact, must save without errors.	High	BR3

4. Non-Functional Requirements

ID	Category	Requirement	Applies To
NFR1	Usability	Messages must be short, specific and shown next to the affected field or as a clear alert.	FR5, FR8, FR9
NFR2	Performance	Validation must run in the browser without noticeable delay and without extra server calls.	All scripts
NFR3	Reliability	Rules must behave the same on every save and on every change of Impact, including reversal.	FR3, FR7
NFR4	Data Integrity	No high-impact incident may be saved with missing ownership or inconsistent Urgency.	FR2, FR5, FR7
NFR5	Maintainability	Each rule lives in a separately named, active configuration record with a clear purpose.	All records
NFR6	Compatibility	Client Scripts must use supported g_form APIs and work with isolated script mode.	FR5, FR7, FR9
NFR7	Security	Configuration changes require administrative or configuration access.	All records
NFR8	Simplicity	The solution must stay lightweight and easy to implement in a short-duration micro project.	All

5. Data Dictionary

The Incident fields used by the solution are listed below.

Label	Field Name	Type	Use in Solution
Impact	impact	Choice	Trigger for the UI Policy and the onChange

Label	Field Name	Type	Use in Solution
			and onSubmit scripts (1 = High, 2 = Medium, 3 = Low).
Urgency	urgency	Choice	Set to 1 by onChange and made read-only by the UI Policy Action.
Assignment group	assignment_group	Reference	Made mandatory by the UI Policy Action.
Assigned to	assigned_to	Reference	Checked by the onSubmit script.
State	state	Choice	Protected from list editing by the onCellEdit script.

6. Configuration Requirements

Record	Table	Type	Key Settings
High Impact Control	Incident	UI Policy	Active, condition Impact is 1 - High, Reverse if false, On load
Assignment group action	Incident	UI Policy Action	Mandatory: True, Read only: False
Urgency action	Incident	UI Policy Action	Read only: True, Visible: Leave alone
Auto set urgency for high impact	Incident	Client Script	onChange, field Impact, Active
Prevent save if Assigned To missing	Incident	Client Script	onSubmit, Active
Prevent state change via list edit	Incident	Client Script	onCellEdit, field State, Active

6.1 Save-Validation Logic

The core of FR7 and FR8 is the onSubmit check below.

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorMessage(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}
```

7. Requirement Workflow

Step	User Action	System Response
1	Open a new Incident form	Form loads; UI Policy evaluates its condition.
2	Set Impact to 1 - High	Assignment group becomes mandatory, Urgency is set to High and becomes read-only, and an information message appears.

Step	User Action	System Response
3	Click Submit with Assigned to empty	Save is blocked and an error appears on Assigned to.
4	Fill in Assigned to and click Submit	Incident saves successfully.
5	Change Impact from High to Medium	Assigned mandatory rule and read-only Urgency are reversed.
6	Double-click State in the list view	Alert appears and the State value remains unchanged.
7	Change State on the form and click Update	State change saves successfully.

8. Assumptions and Constraints

8.1 Assumptions

- The user has administrative or configuration access to a ServiceNow instance.
- The standard Incident table and its Impact, Urgency and State choice values are in use.
- Users work on the desktop UI, where the Client Scripts are configured to run.

8.2 Constraints

- Client-side controls apply only to the browser form and list; they do not cover API, import or server-side updates.
- The solution is limited to the Incident table.
- Validation covers only the rules defined above.

9. Conclusion

The requirements define eleven functional and eight non-functional needs, tied to five business rules. Together they provide a clear basis for the design, planning and development phases that follow.